

Second Degree Price Discrimination

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First Degree

- Firm knows it faces different individuals with different demand functions
- Firm can tell who is who
- Extracts all consumer surplus, usually with a **two-part tariff**

Second Degree

- Firm knows it faces different individuals with different demand functions
- Firm *cannot* tell who is who
- Firm offers a menu of different packages or options
- Designed such that consumers self-select
- Unable to extract all consumer surplus

More specifically...

- Firm knows it faces consumers with different willingness to pay (WTP), but cannot tell who is who
 - ▶ e.g. knows higher income people are willing to pay more, but cannot tell people's income
 - ▶ airlines know business travelers are less flexible and willing to pay more, but cannot tell who are business travelers
- In these cases, firms offer different packages and let consumers **self-select** into different price categories

Consider a monopolist who face N identical high-income consumers, each with inverse demand

$$P_H = A - Q$$

and n identical low-income consumers, each with inverse demand

$$P_L = a - Q$$

with $A > a > 0$. The monopolist has cost function $C = cQ$ with $0 < c < a$.

- Let $W_H(Q)$ denote the WTP of a high-income consumer for Q units

$$W_H(Q) = \int_0^Q P_H(x) dx = AQ - \frac{Q^2}{2}$$

- Let $W_L(Q)$ denote the WTP of a low-income consumer for Q units

$$W_L(Q) = \int_0^Q P_L(x) dx = aQ - \frac{Q^2}{2}$$

- Suppose the monopolist sells packages (Q, V) , consisting of Q unites at a price V
- Then the monopolist has three options:
 - 1 Offer only one type of package (Q, V) bought only by high-income people
 - 2 Offer only one type of package (Q, V) bought by both high-income and low-income people
 - 3 Offer two types of packages (Q_H, V_H) targeted to high-income consumers, and (Q_L, V_L) targeted to low-income consumers

Option 1 - only high-income

- $V > W_L(Q)$, but $V \leq W_H(Q)$
- Monopolist might as well charge $V = W_H(Q)$
- Profit $\pi_1 = N[W_H(Q) - cQ] = N\left(AQ - \frac{Q^2}{2} - cQ\right)$
- $\frac{d\pi_1}{dQ} = 0$ gives $Q_1^* = A - c$, and then:

$$V_1^* = W_H(Q_1^*) = W_H(A - c) = \frac{A^2 - c^2}{2}$$

and

$$\pi_1^* = N[W_H(Q_1^*) - cQ_1^*] = N\frac{(A - c)^2}{2}$$

Option 2 - both high-income and low-income

- $V \leq W_L(Q)$ (which implies $V < W_H(Q)$ since $W_L(Q) < W_H(Q)$)
- Monopolist might as well charge $V = W_L(Q)$
- Profit $\pi_2 = (N + n)[W_L(Q) - cQ] = (N + n) \left(aQ - \frac{Q^2}{2} - cQ \right)$
- $\frac{d\pi_2}{dQ} = 0$ gives $Q_2^* = a - c$, and then (as in the previous slide):

$$V_2^* = W_L(a - c) = \frac{a^2 - c^2}{2}$$

and

$$\pi_2^* = (N + n) \frac{(a - c)^2}{2}$$

Option 3 - two type of packages

(1) $V_L \leq W_L(Q_L)$

L -consumers are willing to buy "their" package

(2) $W_L(Q_L) - V_L \geq W_L(Q_H) - V_H$

Incentive compatibility constraint for L -consumers (they do not prefer the H -package)

(3) $V_H \leq W_H(Q_H)$

H -consumers are willing to buy "their" package

(4) $W_H(Q_H) - V_H \geq W_H(Q_L) - V_L$

Incentive compatibility constraint for H -consumers (they do not prefer the L -package)

- Since for every $Q > 0$, $W_H(Q) > W_L(Q)$, (3) follows from (1) and (4):
 - ▶ from (1) we get $W_L(Q_L) - V_L \geq 0$
 - ▶ using the fact that $W_H(Q_L) > W_L(Q_L)$, we get $W_H(Q_L) - V_L > 0$
 - ▶ by (4), $W_H(Q_H) - V_H > 0$, i.e., $V_H < W_H(Q_H)$
- Thus, from the incentive compatibility constraint for H -consumers we get that H -consumers must be getting a positive surplus
 - ▶ the total price of their package is less than what they are willing to pay for it

Ignore for the moment constraint (2). We will reason as if (2) were not a constraint and then show that it will be satisfied

Profit Maximization

- (1) and (4) must be satisfied as equalities:

$$V_L = W_L(Q_L)$$

$$V_H = W_H(Q_H) - W_H(Q_L) + W_L(Q_L) = AQ_H - \frac{(Q_H)^2}{2} - (A - a)Q_L$$

- The profit function π

$$= N(V_H - cQ_H) + n(V_L - cQ_L)$$

$$= N \left[AQ_H - \frac{(Q_H)^2}{2} - (A - a)Q_L - cQ_H \right] + n \left[aQ_L - \frac{(Q_L)^2}{2} - cQ_L \right]$$

- Solving $\frac{\partial \pi}{\partial Q_H} = N(A - Q_H - c) = 0$ and $\frac{\partial \pi}{\partial Q_L} = -N(A - a) + n(a - Q_L - c) = 0$, we get

$$Q_H^* = A - c, \quad Q_L^* = a - c - \frac{N}{n}(A - a)$$

- Since $A > a$, $Q_H^* > Q_L^*$.
- This solution is acceptable iff $Q_L^* > 0 \iff n(a - c) > N(A - a)$

Check if (2) is also satisfied

- Recall (2) $W_L(Q_L) - V_L \geq W_L(Q_H) - V_H$
- Since (1) is satisfied as an equality, the LHS of (2) is zero
- RHS of (2) is $W_L(Q_H^*) - V_H^* = -(A - a)(Q_H^* - Q_L^*)$ which is negative since $Q_H^* > Q_L^*$

Substituting the optimal quantities in π , we get the maximum profit with second-degree price discrimination is

$$\pi_3^* = \frac{(N+n)(a-c)^2}{2} + \frac{(N+n)N(A-a)^2}{2n} = \pi_2^* + \frac{(N+n)N(A-a)^2}{2n}$$

Thus Option 2 is inferior to Option 3. Firm will only compare Options 1 and 3.

- If $n(a - c) > N(A - a)$, then $\pi_3^* > \pi_1^*$.
- Intuitively, the marginal WTP of an L -consumer is a .
- By introducing a new package with a very small quantity, the monopolist increase its profit from L -consumers by $n(a - c)$
- But this gives the H -consumers a surplus of $(A - a)$, while the current package gives them zero surplus.
- To prevent switching, the price of package must be reduced by $(A - a)$ leading to a loss in revenue of $N(A - a)$
- Hence, introducing the new package is profitable iff $n(a - c) > N(A - a)$

In conclusion, the monopolist will use option 3 if $n(a - c) > N(A - a)$ and option 1 if $n(a - c) < N(A - a)$

Example 1

- $N = n, A = 10, a = 4, c = 2$
- Then $n(a - c) = 2n < N(A - a) = 4n$
- Monopolist will choose option 1
- In fact, $\pi_1^* = 32n, \pi_2^* = 4n$ (and π_3^* is not defined because $Q_L^* = -4$)

Example 2

- $N = n$, $A = 10$, $a = 8$, $c = 2$
- Then $n(a - c) = 6n > N(A - a) = 2n$
- Monopolist will choose option 3
- In fact, $\pi_1^* = 32n$, $\pi_2^* = 36n$, and $\pi_3^* = 40n$
- Note that $Q_H^* = 8$, $Q_L^* = 4$, $V_H^* = 40$ and $V_L^* = 24$.
- effective prices per unit are $P_H = \frac{V_H}{Q_H} = 5$ and $P_L = \frac{V_L}{Q_L} = 6$
- The package designed for H -consumers incorporates a quantity discount.